

AP-E7 on the Actual $SO(3)$ Soliton Orbit: FR Torsion, Pfaffian Holonomy, CPT, and a Minimal Rank-Three Resolution of the $c_1 = x + 2y$ Obstruction

Route F research note

17 July 2026

Abstract

The canonical AP-E6 relaxed charged two-colour $B = 1$ profile has localized collective orbit $SO(3) = SU(2)/\mathbb{Z}_2$, not $\mathbb{CP}^1$. We therefore replace the previously assumed free $\mathcal{O}(2)$ line by the exact classification of lines on the actual orbit. The cellular complex of $\mathbb{RP}^3$ gives

$$H^2(SO(3); \mathbb{Z}) = \mathbb{Z}_2, \quad t = \beta(a), \quad a \in H^1(SO(3); \mathbb{Z}_2),$$

and the unique nontrivial flat line is $\mathcal{L}_{\text{FR}} = SU(2) \times_{\text{sign}} \mathbb{C}$. Its generator-loop holonomy is -1 , its square is trivial, and its real form is the candidate FR Pfaffian line. A determinant obtained by squaring that real Pfaffian has holonomy $+1$. However, the usual QCD WZW/FR rule gives $(-1)^{N_c B} = +1$ for $N_c = 2, B = 1$, conditionally selecting the trivial bosonic sector. The actual Yukawa mapping-torus mod-two index has not been computed with a common microscopic regulator, so this standard selection is not promoted to a same-operator Pfaffian theorem.

The torsion line is the correct alternative carrier on $SO(3)$, but it is not equivalent to $\mathcal{O}(2)$: its de Rham curvature vanishes and $2t = 0$, whereas $c_1(\mathcal{O}(2))$ is a nonzero free class. Their collective spectra also disagree. CPT inversion preserves both torsion classes. We construct one deck-compatible topological Real lift, while keeping the physical CPT operator, its square, spectral cut, and Pfaffian orientation open.

In parallel we solve the earlier two-band obstruction at the level of bundle mathematics. On $X = \mathbb{CP}_z^1 \times \mathbb{CP}_w^1$, the three sections

$$s = (z_0 w_0^2, z_1 w_1^2, z_0 w_1^2 + z_1 w_0^2)$$

of $\mathcal{O}(1, 2)$ have no common zero. With $P_+ = \bar{s} s^\dagger / \|s\|^2$,

$$\mathcal{M} = m(2P_+ - \mathbf{1}_3), \quad \text{spec } \mathcal{M} = \{+m, -m, -m\}, \quad c_1(E_+) = x + 2y.$$

The negative rank-two complement has $c_1(E_-) = -x - 2y$ and $c_2(E_-) = 4xy$, so $E_+ \oplus E_- = X \times \mathbb{C}^3$ exactly. This proves that rank three is enough, and is minimal relative to the two-band obstruction. It remains a spectator: the $\mathbb{CP}^1$ factor is not a normalizable modulus of the canonical soliton and no charged two-colour Yukawa embedding has produced these three bands. Hence every physical closure and portal gate remains false.

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1 Scope, provenance, and theorem-level answer

This note addresses two questions left by AP-E6.

- (i) What determinant or Pfaffian line can exist on the actual localized moduli $SO(3)$, and what does CPT do to it?
- (ii) Is the obstruction $c_1^2 \neq 0$ intrinsic, or only a defect of the two-band asymptotic mass?

The first question is answered exactly at the topological level and fail-closed at the microscopic-regulator level. The second has an explicit rank-three success, fail-closed as a physical same-soliton construction.

The verifier imports the public AP-E6 Workline-B solver, calls

```
solve_relaxed_profile(box=16.0,sample_points=4097),
```

and hard-asserts equality among its recomputed little-endian float64 (r, F, s) digest and the Workline-B and Workline-C machine cards:

```
81104f59a5f3f4337739fc2a217cafc8da91581c9f1fb40b4fdb6ce0f948d59.
```

The bound solution has $B = 0.999999999686391$ and $s(0) = 0.917128964607058$. Thus the topology below is not attached to a separately solved or idealized hedgehog.

Theorem 1.1 (Strongest defensible AP-E7 result). *Let $\mathcal{Q} = SO(3)$ be the localized orbit of the canonical AP-E6 $B = 1$ profile. Then:*

- (a) *complex line bundles on $\mathcal{Q}$ form $\mathbb{Z}_2$; the nontrivial class is the flat FR sign line and has generator holonomy -1 ;*
- (b) *its real sign form can be a Pfaffian line, but the actual family chooses it only if its mapping-torus mod-two index is odd;*
- (c) *inversion preserves the class and admits a topological Real lift, but topology does not determine the microscopic CPT regulator;*
- (d) *no line on $\mathcal{Q}$ is equivalent to the free $\mathcal{O}(2)$ line on $\mathbb{CP}^1$;*
- (e) *a globally gapped rank-three Hermitian mass on the spectator $\mathbb{CP}^1 \times \mathbb{CP}^1$ realizes $c_1(E_+) = x + 2y$ exactly.*

Nevertheless,

```
actual_pfaffian_torsion_class_selected=false,
microscopic_cpt_regulator_constructed=false,
physical_rank_three_yukawa_embedding_derived=false,
lane_closed=false,      portal_start_allowed=false.
```

The determinant-line and eta-holonomy distinction follows the families framework of Bismut–Freed and Dai–Freed [1, 2]. FR signs are configuration-space quantization data rather than a consequence of merely writing a soliton ansatz [3, 4].

2 Exact topology of the physical orbit

2.1 Cellular computation of $H^2(SO(3); \mathbb{Z})$

Use $SO(3) \simeq \mathbb{RP}^3$, with one cell in each dimension 0, 1, 2, 3. For the usual cellular orientations the integral boundary is

$$d_k = 1 + (-1)^k, \quad 0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0.$$

Dualizing gives

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0,$$

where the displayed terms now have cohomological degrees 0, 1, 2, 3. Consequently

$$H^0 = \mathbb{Z}, \quad H^1 = 0, \quad H^2 = \mathbb{Z}/2, \quad H^3 = \mathbb{Z}. \quad (1)$$

In particular $H_{\text{dR}}^2(SO(3)) = 0$. There is no continuous magnetic flux or free Chern-number parameter on this orbit.

2.2 The Bockstein and the unique nontrivial line

Let a generate $H^1(\mathbb{RP}^3; \mathbb{Z}_2) = \mathbb{Z}_2$, and use the coefficient sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}_2 \longrightarrow 0.$$

Represent a by the mod-two cellular cochain 1, lift it to the integer cochain $A = 1$, and note that $\delta A = 2$. Dividing by two before taking the integral class gives

$$\beta(a) = \left[\frac{\delta A}{2} \right] = [1] = t, \quad H^2(\mathbb{RP}^3; \mathbb{Z}) = \{0, t\}, \quad 2t = 0. \quad (2)$$

Since complex line bundles are classified by $H^2(-; \mathbb{Z})$, these are the only two topological possibilities.

The universal cover is

$$p : SU(2) = S^3 \longrightarrow SO(3) = \mathbb{RP}^3, \quad A \sim -A.$$

The nontrivial line has the concrete associated-bundle model

$$\mathcal{L}_{\text{FR}} = SU(2) \times_{\mathbb{Z}_2, \text{sign}} \mathbb{C}, \quad (A, z) \sim (-A, -z), \quad (3)$$

so a section lifted to $SU(2)$ obeys $\psi(-A) = -\psi(A)$. Its pullback to S^3 is trivial because $H^2(S^3; \mathbb{Z}) = 0$, but its descent data are not. The real sign line

$$\mathcal{P}_{\text{FR}} = SU(2) \times_{\text{sign}} \mathbb{R}$$

has $w_1(\mathcal{P}_{\text{FR}}) = a$, and $\mathcal{P}_{\text{FR}} \otimes_{\mathbb{R}} \mathbb{C} = \mathcal{L}_{\text{FR}}$ has $c_1 = \beta(a) = t$.

2.3 Generator holonomy and rotor sectors

A generator of $\pi_1(SO(3)) = \mathbb{Z}_2$ lifts to

$$A(\tau) = \begin{pmatrix} e^{i\pi\tau} & 0 \\ 0 & e^{-i\pi\tau} \end{pmatrix}, \quad 0 \leq \tau \leq 1, \quad A(1) = -\mathbf{1}_2. \quad (4)$$

Closing the lift uses the sign transition in (3); hence

$$\text{Hol}_\gamma(\mathcal{L}_{\text{FR}}) = -1, \quad \text{Hol}_\gamma(\mathcal{L}_{\text{FR}}^{\otimes 2}) = +1. \quad (5)$$

Peter–Weyl functions in spin j transform under the centre as $(-1)^{2j}$. The trivial sector therefore contains integer j , while the FR sign sector contains half-integer j . The lowest full rigid-rotor block in the odd sector is $j = 1/2$ and has dimension $(2j + 1)^2 = 4$, not three. Additional hedgehog constraints may reduce the physical spin/isospin labeling, but they cannot turn the torsion line into the $\mathbb{CP}^1$ holomorphic space $H^0(\mathbb{CP}^1, \mathcal{O}(2))$.

3 Determinant, Pfaffian, and the missing mapping-torus calculation

3.1 What topology determines

For a real skew-adjoint Fredholm family with Pfaffian line $\text{Pf} \rightarrow \mathcal{Q}$, the loop sign is controlled by mod-two spectral flow, equivalently by the appropriate mapping-torus mod-two index:

$$\text{Hol}_\gamma(\text{Pf}) = (-1)^{\text{ind}_2(\mathcal{D}_{Y_\gamma})}. \quad (6)$$

On $SO(3)$, equation (6) leaves exactly two options: the trivial real line if the index is even, and $\mathcal{P}_{\text{FR}}$ if it is odd. For the standard doubled skew matrix, the determinant line is the Pfaffian square,

$$\text{Det} \simeq \text{Pf}^{\otimes 2}, \quad \text{Hol}_\gamma(\text{Det}) = +1 \quad (7)$$

in either torsion sector. This does not say that every complex chiral determinant is automatically the square of one real Pfaffian; the real structure and representation must first be specified. A single pseudoreal $SU(2)$ Weyl doublet can carry the familiar mod-two anomaly, whereas vectorlike doubling gives an even representation-parity control [5].

3.2 Conditional two-colour selection

The standard Skyrmion WZW/FR rotation rule gives

$$\text{Hol}_{2\pi} = (-1)^{N_c B}. \quad (8)$$

For the present two-colour, unit-charge target,

$$N_c = 2, \quad B = 1, \quad (-1)^{N_c B} = +1. \quad (9)$$

Thus, *if* the usual WZW derivation is realized by the same charged two-colour regulator, it selects the trivial bosonic line, as expected for a two-colour diquark baryon. The AP-E6 declared constituent Dirac completion also contains two Weyl doublets and has even mod-two representation parity. These are strong consistency controls, not the requested operator theorem: the common Euclidean/PV operator, its mapping-torus domain, eta prescription, and Pfaffian orientation are still absent. WZW quantization and global current algebra supply the context for (8) [6].

Fail-closed gate 3.1 (Actual Pfaffian class). The machine card sets

```
actual_yukawa_mapping_torus_mod2_index_computed=false
actual_pfaffian_torsion_class_selected=false.
```

The conditional value +1 in (9) is not silently substituted for a same-operator APS/PV computation.

4 CPT on the torsion line

Use the base involution

$$\iota : SO(3) \longrightarrow SO(3), \quad \iota([A]) = [A^{-1}]. \quad (10)$$

Since inversion acts trivially on $\pi_1(SO(3)) = \mathbb{Z}_2$, it preserves both flat characters. Equivalently,

$$c_1(\overline{\mathcal{L}}_{\text{FR}}) = -t = t,$$

because $2t = 0$. One explicit topological anti-linear lift is

$$J[A, z] = [A^{-1}, \bar{z}]. \quad (11)$$

It is well-defined: replacing (A, z) by $(-A, -z)$ sends the right-hand side to $[-A^{-1}, -\bar{z}] = [A^{-1}, \bar{z}]$. This particular lift satisfies $J^2 = 1$.

Existence is not uniqueness of the microscopic CPT operation. A physical antiunitary map must also act on spin, gauge, flavour and regulator spaces; preserve or transform the operator domain and boundary conditions; choose a spectral cut; and transport the determinant/Pfaffian orientation. Its square on the full Hilbert space can contain fermion parity or internal symmetries not seen by the one-dimensional topological carrier. We therefore record (11) as a Real-line construction and keep `microscopic_cpt_regulator_constructed=false`.

5 Why torsion cannot replace $\mathcal{O}(2)$

Let $u = c_1(\mathcal{O}(1))$ generate $H^2(\mathbb{CP}^1; \mathbb{Z}) = \mathbb{Z}$. The old carrier has

$$c_1(\mathcal{O}(2)) = 2u \neq 0,$$

a free class visible in de Rham cohomology and Berry curvature. On the actual orbit, every line has class 0 or t , with

$$2t = 0, \quad t \otimes \mathbb{R} = 0, \quad F_{\nabla_{\text{flat}}} = 0.$$

There is no class-preserving identification $2u \leftrightarrow t$. More physically, $\mathcal{O}(2)$ produces three Dolbeault sections on $\mathbb{CP}^1$, while the FR line imposes a centre parity on an $SO(3)$ rotor. Therefore the torsion line is the correct *classification replacement* on the true moduli but not an equivalent realization of the AP-E4 three-state mechanism.

6 Explicit minimal rank-three mass family

6.1 A basepoint-free map to $\mathbb{CP}^2$

Let

$$X = \mathbb{CP}_z^1 \times \mathbb{CP}_w^1, \quad x = c_1(\mathcal{O}(1, 0)), \quad y = c_1(\mathcal{O}(0, 1)),$$

so $x^2 = y^2 = 0$ and $\int_X xy = 1$. Consider

$$\begin{aligned} s_0 &= z_0 w_0^2, \\ s_1 &= z_1 w_1^2, \\ s_2 &= z_0 w_1^2 + z_1 w_0^2. \end{aligned} \quad (12)$$

All three are sections of $\mathcal{O}(1, 2)$.

Lemma 6.1 (No common zero). *The sections in (12) are basepoint-free.*

Proof. If $s_0 = s_1 = 0$, then $(z_0 = 0 \text{ or } w_0 = 0)$ and $(z_1 = 0 \text{ or } w_1 = 0)$. The branches $z_0 = z_1 = 0$ and $w_0 = w_1 = 0$ are forbidden in projective space. In the remaining branch $z_0 = 0, w_1 = 0$, one has $z_1 w_0^2 \neq 0$, so $s_2 \neq 0$. In the branch $w_0 = 0, z_1 = 0$, one has $z_0 w_1^2 \neq 0$, again giving $s_2 \neq 0$. Thus no common zero exists. $\square$

The normalized holomorphic column $v = s/\|s\|$ defines a map $f : X \rightarrow \mathbb{CP}^2$. Its span is the pulled-back tautological line

$$\text{span}(v) = f^* \mathcal{O}_{\mathbb{CP}^2}(-1), \quad c_1(\text{span}(v)) = -(x + 2y).$$

This fixes the sign that a bare “Veronese vector” can easily obscure. Complex conjugation reverses the Chern class. Hence

$$e_+ = \frac{\bar{s}}{\|s\|}, \quad P_+ = e_+ e_+^\dagger, \quad c_1(E_+) = x + 2y. \quad (13)$$

6.2 Hermitian mass and uniform gap

For any constant $m > 0$, define

$$\mathcal{M}(z, w) = m(2P_+(z, w) - \mathbf{1}_3). \quad (14)$$

Because $P_+^2 = P_+ = P_+^\dagger$ globally,

$$\text{spec } \mathcal{M} = \{+m, -m, -m\}, \quad \mathcal{M}^2 = m^2 \mathbf{1}_3. \quad (15)$$

Thus the gap to zero is exactly m and the positive–negative band separation is $2m$, uniformly on all of X . Compactness plus the basepoint-free lemma ensures the projector has no singular normalization.

6.3 Complement bundle and K-theory identity

Let $E_- = E_+^\perp$, a rank-two bundle. Since the ambient three-band bundle is trivial,

$$E_+ \oplus E_- = X \times \mathbb{C}^3, \quad c(E_-) = (1 + c_1(E_+))^{-1}. \quad (16)$$

Writing $\ell = x + 2y$ and using $\ell^2 = 4xy$,

$$c_1(E_-) = -\ell = -(x + 2y), \quad c_2(E_-) = \ell^2 = 4xy. \quad (17)$$

Equivalently in K-theory and rational cohomology

$$[E_-] = 3 - [E_+], \quad \text{ch}(E_-) = 3 - e^\ell = 2 - \ell - \frac{\ell^2}{2} = 2 - (x + 2y) - 2xy. \quad (18)$$

These identities are the precise cancellation that a total trivial rank-three bundle requires; no nonexistent “complement line” is introduced. Standard bundle classification and K-theory provide the general framework [7, 8].

6.4 Minimality relative to the two-band no-go

A gapped 2×2 Hermitian mass with one positive and one negative band defines a map to the Grassmannian $\text{Gr}_1(\mathbb{C}^2) = \mathbb{CP}^1$. If $u \in H^2(\mathbb{CP}^1; \mathbb{Z})$, then $u^2 = 0$, so every positive eigenline pulled back from this target obeys

$$c_1(E_+)^2 = 0.$$

The requested class obeys

$$(x + 2y)^2 = 4xy \neq 0.$$

For three bands the target is $\mathbb{CP}^2$, whose hyperplane class has nonzero square. Equations (13)–(15) therefore show both sufficiency and minimality: rank three is the first nontrivial sign-split band number that can realize the class in a trivial ambient bundle.

7 Numerical and symbolic verification

The deterministic verifier performs the following independent controls.

- (1) It evaluates the $\mathbb{RP}^3$ cellular boundaries $(0, 2, 0)$, the Bockstein lift $\delta A/2 = 1$, and the FR character square.
- (2) It samples the path (4) at 65 points and checks unitarity, determinant one, endpoints, inversion, and deck compatibility.
- (3) It scans 9×12 points on each $\mathbb{CP}^1$, or 11,664 points on X , at $m = 1.7$. The sampled minimum is $\|s\|^2 = 0.25$; projector, Hermiticity, and eigenvalue residuals are at floating-point roundoff, and the gap is 1.7.
- (4) Gauge-invariant triangulated Berry sums with 1,520 triangles per sphere give

$$\frac{i}{2\pi} \int_{\mathbb{CP}_z^1} \text{Tr}(P_+ dP_+ \wedge dP_+) = 1, \quad \frac{i}{2\pi} \int_{\mathbb{CP}_w^1} \text{Tr}(P_+ dP_+ \wedge dP_+) = 2$$

to better than 10^{-11} .

The matrix gap is an exact finite-band theorem. It is not the Fredholm gap of the spatial Yukawa–Callias operator. General Callias theory requires the actual asymptotic potential and operator class, not merely a target bundle template [9].

Table 1: AP-E7 evidence boundary.

Statement	Status
$H^2(SO(3); \mathbb{Z}) = \mathbb{Z}_2$, FR cover and both flat sectors	exact
FR sign-line and Pfaffian-candidate holonomy -1	exact topology
Standard $(-1)^{N_c B} = +1$ at $N_c = 2, B = 1$	conditional UV rule
Actual Yukawa mapping-torus mod-two index	open
Deck-compatible topological CPT Real lift	constructed
Microscopic CPT/PV regulator and Pfaffian orientation	open
Rank-three $c_1(E_+) = x + 2y$ mass and uniform matrix gap	exact spectator
Same-soliton rank-three Yukawa embedding and Fredholm gap	open
Degree-one Route-E portal	forbidden

8 Blockers and next discriminating calculations

The result narrows the remaining work to operator-level questions.

- B1.** Construct the common Euclidean/PV Yukawa operator along the nontrivial $SO(3)$ loop and evaluate its mapping-torus ind_2 or eta phase. This decides between the two classified Pfaffian lines.
- B2.** Define the regulator CPT action on its domain, spectral cut and orientation. Then compare its transported Real structure with equation (11).
- B3.** Derive three asymptotic fermion bands from an actual charged two-colour representation and match their normalized mass projector to equation (13), including gauge covariance and heavy-threshold decoupling.

- B4.** Replace the spectator $\mathbb{CP}^1$ by a localized normalizable physical modulus, or formulate the family over the actual $SO(3)$ orbit. Without this step the rank-three success cannot close the same-soliton lane.
- B5.** Prove a uniform Fredholm gap for that spatial operator. Only then may the determinant/Pfaffian family be paired with the WZW descent and tested as a portal antecedent.

References

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